

# Amalraj J.C.Varatharajan

## ELECTRICAL/INFORMATIC ENGINEER

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📍 Chanteloup-les-Vignes,  
France

ENSEA graduated engineer specializing in **embedded systems**, with strong multi-language programming skills on both Windows and Linux. Curious and innovation-driven, I am looking for **ambitious projects** to design efficient and high-performance solutions.

### EDUCATION

#### ENSEA, Cergy — Apprenticeship Engineering Degree

Sept. 2022 – Sept. 2025

- General engineering school specializing in **electronics**, **computer science**, and **networks**.
- Specialized tracks: **Embedded Systems**, and Identification & Security
- Summer Program 2023 — Barcelona, Spain, TBS Education: Introduction to **Machine Learning**, **Linux**, and **Cybersecurity**
- Summer Program 2024 — Seoul, South Korea, Hanyang University: Introduction to **Android Application Development**

### EXPERIENCE

#### Apprentice Engineer — NKAD PRINTERS, Buchelay/Paris

Dec. 2022 – Sept. 2025

- **Python** program for **analyzing metal 3D printing temperatures** from a GCode file
- Manufacturing and **programming of drones** using **Arduino/Pixhawk**, including component sourcing and assembly
- Development of a **website** hosted on a **Raspberry Pi** : **Python**, **HTML**, and **PHP**
- **Programming** and development on a **Zynq UltraScale+ MPSoC** board: **Vivado**, **Vitis**, **PetaLinux**
- **Proof of Concept** for an **embedded system** for metal 3D printing: **AI-based supervision** and defect detection

#### Developer Intern — ECAM-EPMI, Cergy

Avr. 2022 – Jun. 2022

- Project: development of a **brain signal analysis program in Python**
- Contribution to a research paper on Wavelets Statistical Denoising (WaSDe): individual evoked potential extraction via multi-resolution wavelet decomposition and bootstrap

### SKILLS

#### Programming :

- Languages: C/C++, Python, Java, HTML/XML
- Web and Android app development
- Experience with STM32, Raspberry Pi, Linux

#### FPGA & Embedded Systems:

- Tools: AMD Vivado, Vitis, Vitis HLS, PetaLinux
- Language: VHDL
- SoC FPGA design and real-time systems

#### Electronics :

- Robotics
- Electronic schematics and circuits (drones, IR remote controls)

#### AI & Machine Learning:

- Knowledge of various neural network architectures
- Implementation of machine learning on embedded systems

#### Soft Skills :

- Project management
- Teamwork
- Autonomy
- Adaptability

### PROJECTS

#### Proof of Concept — Embedded System for Metal 3D Printing

Jan. 2025 – Sept. 2025

- Development on a **Zynq UltraScale + MPSoC** board
- System capable of **supervision** and **object recognition** in images using an **embedded AI**

#### Karting GPS Tracker

Feb. 2025 – Jun. 2025

- Development of an **interactive Python application**
- GPS tracking, weather display via **API**, lap time monitoring

#### Identification & Security System

Mar. 2024 – Jun. 2024

- Face recognition and RFID badge identification system
- HuskyLens camera, RFID reader, and **Raspberry Pi**

### LANGUES

#### French

Native

#### English

TOEIC : 940  
B2-C1 Level

#### Spanish

B1 Level